# Awadhesh K. Das

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## Research Summary

- Experimental condensed-matter physicist specializing in **nonlinear optical spectroscopy**, **ultrafast pump-probe** techniques, and symmetry-sensitive **rotational anisotropy nonlinear harmonic generation (RA-NHG)** for investigating emergent electronic and structural phases in quantum materials.
- Experienced in designing, building, aligning, and troubleshooting multi-stage optical setups, performing cryogenic measurements, and automating data acquisition using LabVIEW and NI hardware.
- Experienced in nonlinear susceptibility tensor analysis, quantitative data modeling, and interpreting optical responses to investigate **charge density wave (CDW) and hidden-order** phenomena in correlated and topological materials.

## SKILLS & EXPERTISE

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| **Laser/Optics:** | Maintenance \| Troubleshooting \| Ti:sapphire femtosecond laser \| Regenerative amplifier \| Dual channel OPA \| Beam alignment \| polarization optics |
| **Ultrafast spectroscopy:** | RA-NHG \| Pump-Probe spectroscopy \| Transient reflectivity \| Time-resolved Kerr effect \| Time-domain THz emission spectroscopy \| Photogalvanic effects (LPGE/CPGE) \| cryogenic measurements |
| **Instrumentation:** | MATLAB \| Mathematica \| LabVIEW \| NI DAQ \| Zaber stages \| Lock-in amplifiers \| Optical choppers \| delay stages synchronization \| Cryogenic instrumentation |
| **Scientific tools:** | LaTeX \| Adobe Illustrator \| VESTA \| Excel \| PowerPoint \| Microsoft Word |
| **Materials and characterization:** | Chemical vapor deposition \| Spin coating \| Dry etching \| Atomic force microscopy \| Raman spectroscopy \| X-ray crystallography |

## Research Experience

**Graduate Researcher: Temple University** | Philadelphia, PA Jan 2024–Present

**Project 1: Charge density wave ordering on the surface of the Weyl semimetal CoSi.**

- Performed temperature-dependent **RA-SHG** measurements to investigate electronic ordering on the (001) surface of CoSi.
- Identified a **surface-confined CDW** ordering below $\sim 90$ K, with no corresponding bulk transition.
- Extracted **critical behavior** of the phase transition through nonlinear susceptibility analysis and power-law fitting.

**Project 2: Hidden-order phase in the correlated heavy fermion metal $URu_2Si_2$.**

- Performed **temperature-dependent RA-THG** measurements to probe the nonlinear optical response across the hidden-order transition.
- Resolved a distinct nonlinear optical signature emerging at the hidden-order transition temperature $\sim 17.5$ K, establishing THG as a **sensitive probe** of the hidden-order phase.
- Observed **Ising-like behavior** in the temperature-dependent THG intensity response and extracted **critical behavior** near the hidden-order transition.

**Collaborative Research: Broadband optical pump–probe measurements in the Weyl semimetal RhSi.**

- Performed **polarization- and fluence-dependent transient reflectivity** measurements with wavelength-resolved probe detection in the Weyl semimetal RhSi.
- Analyzed transient optical spectra to extract spectral signatures of photoinduced **carrier relaxation dynamics** and **coherent phonon amplitudes**.

**Collaborative Research: $f$-band conduction electron hybridization using SHG response of LnAlSi.**

- Investigated **$f$-electron states and conduction electron hybridization** in the LnAlSi (Ln = La, Ce, Pr, Nd) Weyl semimetal series.
- Revealed **Fano interference** arising from coupling between electric- and magnetic-dipole nonlinear optical processes.
- Modeled SHG polarimetry data to track the evolution of nonlinear optical **ellipticity** across the lanthanide series.

## Teaching Experience

**Graduate Teaching Assistant: Temple University** | Philadelphia, PA Aug 2021–Dec 2023

- Instructed and evaluated performances of over 350 students in PHYS 1021/1022 (Introduction to General Physics I & II), PHYS 1061 (Elementary Classical Physics I), PHYS 2022 (General Physics II), and PHYS 0834 (Exploring the Cosmos).

## Publications

1. **A. K. Das**, W. E. Deeg, S. Subedi, C. Shekhar, C. Felser, and D. H. Torchinsky, "Second Harmonic Generation Spectroscopy of the Surface Charge Density Wave in the Weyl Semimetal CoSi", arXiv:2608.17303 (2026).
2. J. Qianni, E. D. Roberts, B. Gonzalez, **A. Das**, D. H. Torchinsky, T. Birol, R. M. Fernandes, and I. R. Fisher, "Evidence for Ferroaxial Order in 1T-$TiSe_2$ via Elastoresistivity Measurements", arXiv:2603.14613 (2026).

*In Preparation:*

3. M. Rai, S. Subedi, **A. K. Das**, B. Lu, K. Wagle, K. Manna, C. Felser, J. Perdew, D. H. Torchinsky, "Broadband Optical Pump-Probe Spectroscopy of RhSi: Band-Selective Coherent Phonon Dynamics and Electronic Relaxation in a Multifold Semimetal", (2026). (Submission by the end of this month)
4. **A. K. Das**, S. Subedi, M. Zdilla, M. B. Maple, and D. H. Torchinsky, "Unveiling the Nature of Hidden Order in the Heavy Fermion Compound $URu_2Si_2$ Using Third Harmonic Generation Spectroscopy", (2026).
5. S. Subedi*, **A. K. Das***, X. Yao, B. Lu, F. Tafti, and D. H. Torchinsky, "Strong $f$-Band Conduction Electron Hybridization Observed Using Second Harmonic Generation Response of the LnAlSi (Ln = La, Ce, Pr, and Nd) Series of Weyl Semimetals", (2026). (*Equal contribution)

## Presentations & Colloquia

- **Second Harmonic Generation Spectroscopy of the Charge Density Wave in the Weyl Semimetal CoSi** | *APS Global Physics Summit* (2026)
- **Nonlinear Optical Characterization of Kagome Lattice Structure** | *Temple Graduate Symposium* (2024)
- **Synthesis and Characterization of 1T′-$WTe_2$ Thin Films** | *Temple Physics Research Symposium* (2022)

## Education

**Ph.D. in Physics**: Temple University | Philadelphia, PA Fall 2026 (Expected graduation)
*Dissertation:* ***"Symmetry-Sensitive Nonlinear Optical Spectroscopy of Emergent Electronic Orders in Quantum Materials"***
Advisor: Prof. Darius H. Torchinsky

## References

1. Prof. Darius H. Torchinksy | Department of Physics, Temple University
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2. Prof. Robert Stanley | Department of Chemistry, Temple University
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3. Prof. Peter Riseborough | Department of Physics, Temple University
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